The CPP Ground Set and the (S, U, π) Framework

Robin Langer

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Abstract

We reformulate the ground set for the sign-reversing involution on $Q_{n,c}(q)$. The previous framework used pairs $(S, \mathfrak{d})$ where $\mathfrak{d}$ is an ascending list of cylindric diagrams (ALCD), which requires the polynomials $h_m(q) = (q)_m \cdot g_m(q)$ to have nonneg coefficients. Since this is not guaranteed in general, we replace the ALCD by a genuine cylindric plane partition π and absorb the signs from $(q)_m$ into an auxiliary subset U. The resulting ground set consists of triples (S, U, π) and does not require any nonnegativity assumption beyond the obvious $g_m \geq 0$. We describe the lattice path interpretation: π is always a family of non-intersecting paths on the cylinder, while S and U contribute two independent layers of alternating signs.

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1 Motivation: why h_m is not enough

Recall the q-binomial decomposition

$$Q_{n,c}(q) = \sum_{j=0}^{n} (-1)^j \, q^{\binom{j+1}{2}} \begin{bmatrix} n \\ j \end{bmatrix}_q h_{n-j}(q), \tag{1}$$

where $h_m(q) = (q;q)_m \cdot g_m(q)$ and $g_m(q) = [z^m]\ \mathrm{GK}_c(z,q)$ is the generating function for cylindric plane partitions of height m.

In the previous framework (CANONICAL_PEEL.tex), the ground set was

$$\mathcal{G}_{n,c} = \big\{(S, \mathfrak{d}) : S \subseteq [n],\ \mathfrak{d} \in H_{n-|S|,c}\big\},$$

where $H_{m,c}$ is a combinatorial set of cardinality $h_m(q)$ (counted with q-weight). This requires every coefficient of $h_m(q)$ to be nonneg.

Remark 1. Computationally, $h_m(q)$ is nonneg for all profiles tested with $d \not\equiv 0 \pmod 3$ and $d \leq 8$. However, for $d \equiv 0 \pmod 3$ it *is* negative.

Example 2 ($d = 3$: h_1 has negative coefficients)**.** For $c = (2,1,0)$, $d = 3$, the CPP count at height 1 is

$$g_1(q) = 2q + 3q^2 + 3q^3 + 3q^4 + 4q^5 + 3q^6 + 3q^7 + 4q^8 + 3q^9 + 3q^{10} + 4q^{11} + \cdots$$

The coefficients oscillate with period 3: $\ldots, 4, 3, 3, 4, 3, 3, \ldots$. Since $h_1 = (1-q) \cdot g_1$, we get

$$h_1(q) = 2q + q^2 + q^5 - q^6 + q^8 - q^9 + q^{11} - q^{12} + \cdots$$

The coefficient of q^k in h_1 is $g_1[k] - g_1[k-1]$, which is negative whenever g_1 drops (e.g. $g_1[6] = 3 < 4 = g_1[5]$, giving $h_1[6] = -1$).

This means the ALCD ground set $\{(S, \mathfrak{d})\}$ has -1 elements at weight 6 — not a genuine set.

The fix is to go back to the *always-nonneg* quantity $g_m(q)$, which literally counts CPPs, and to absorb the alternating signs from $(q)_m$ into a separate subset.

2 The new ground set

We start from the identity

$$Q_{n,c}(q) = \sum_{j=0}^{n} (-1)^j \, q^{\binom{j+1}{2}} \cdot \frac{(q;q)_n}{(q;q)_j} \cdot g_{n-j}(q). \tag{2}$$

This is obtained from (1) by writing $\begin{bmatrix} n \\ j \end{bmatrix}_q \cdot (q;q)_{n-j} = (q;q)_n/(q;q)_j$ and noting $h_{n-j} = (q;q)_{n-j} \cdot g_{n-j}$.

We decompose each factor combinatorially.

Factor 1: the Euler part. The term $(-1)^j q^{\binom{j+1}{2}}$ comes from the initial segment $S = \{1, 2, \ldots, j\}$:

$$(-1)^j \, q^{1+2+\cdots+j} = (-1)^{|S|} \, q^{\sigma(S)}, \qquad S = [j]. \tag{3}$$

Factor 2: the $(q)_n/(q)_j$ part.

$$\frac{(q;q)_n}{(q;q)_j} = \prod_{k=j+1}^{n} (1 - q^k) = \sum_{U \subseteq \{j+1,\ldots,n\}} (-1)^{|U|} \, q^{\sigma(U)}. \tag{4}$$

Factor 3: the CPP part.

$$g_{n-j}(q) = \sum_{\pi \in \mathrm{CPP}_{n-j,c}} q^{\mathrm{wt}(\pi)}, \tag{5}$$

where $\mathrm{CPP}_{m,c}$ denotes the set of rank-r cylindric plane partitions of profile c and height m.

Definition 3 (The ground set $\mathcal{G}_{n,c}$)**.** The *ground set* is

$$\mathcal{G}_{n,c} = \Big\{ (S,U,\pi) : \begin{array}{l} S = \{1,\dots,j\} \text{ for some } 0 \le j \le n, \\ U \subseteq \{j+1,\dots,n\}, \\ \pi \in \mathrm{CPP}_{n-j,\,c} \end{array} \Big\}.$$

Each element carries:

- **Sign:** $\varepsilon(S,U,\pi) = (-1)^{|S|+|U|}$.
- **Weight:** $w(S,U,\pi) = \sigma(S) + \sigma(U) + \mathrm{wt}(\pi)$.

Proposition 4.

$$Q_{n,c}(q) = \sum_{(S,U,\pi)\in\mathcal{G}_{n,c}} (-1)^{|S|+|U|}\, q^{\sigma(S)+\sigma(U)+\mathrm{wt}(\pi)}.$$

Proof. Expand (2) using (3)–(5). □

3 The combined set $T = S \cup U$

Set $T = S \cup U \subseteq [n]$. Since $S = [j]$ and $U \subseteq \{j+1,\dots,n\}$, the sets S and U are disjoint and we have

$$|T| = |S| + |U| = j + |U|, \qquad \sigma(T) = \sigma(S) + \sigma(U).$$

So the sign and weight simplify:

$$\varepsilon = (-1)^{|T|}, \qquad w = \sigma(T) + \mathrm{wt}(\pi).$$

Remark 5. The decomposition $T = S \cup U$ is *not* unique: a subset $T \subseteq [n]$ can arise from different values of j. Specifically, j can be any value in $\{0,1,\dots,n\}$ such that $[j] \subseteq T$. The maximum such j is the length of the initial consecutive run in T:

$$j_{\max}(T) = \max\{k \ge 0 : \{1,\dots,k\} \subseteq T\}.$$

Different choices of j give different CPP heights $n - j$, so the ground set is genuinely *three-dimensional*: the triple (S,U,π) carries more information than the pair (T,π).

4 Lattice path interpretation

Each component of the triple (S,U,π) has a clear role in the lattice path picture.

The CPP π: non-intersecting paths

A cylindric plane partition $\pi = (\nu^0, \nu^1, \ldots, \nu^{r-1})$ of profile c and height m encodes a family of r non-intersecting lattice paths on the cylinder with m levels. Each level corresponds to an entry value $1 \le s \le m$. The interlacing conditions

$$\nu_p^i \ge \nu_{p+c_i}^{i+1} \qquad \text{(indices mod } r\text{)}$$

ensure that the paths do not cross.

Key point: π is always a family of **non-intersecting** paths. There are no "intersecting CPPs" in this framework.

$S = \{1, \ldots, j\}$: peeled layers

The initial segment $S = [j]$ records that the bottom j layers of a height-n path family have been *peeled off*. The CPP π has height $n-j$ (it lives on the *inner* cylinder after removing the j outermost layers).

Geometrically: a height-n CPP has layers $1, 2, \ldots, n$. Layer s is the horizontal strip between the $(s-1)$-th and s-th level. Peeling layers $1, \ldots, j$ means these layers are "detached" from the CPP and contribute only their level numbers $1 + 2 + \cdots + j = \sigma(S)$ to the weight.

The sign $(-1)^j$ reflects that each peeled layer introduces one "virtual crossing" between the detached paths and the remaining family.

$U \subseteq \{j+1, \ldots, n\}$: base removal

In the E–H factorisation $\mathrm{GK}_c = F_c/(zq;q)_\infty$, the factor $1/(q;q)_m = \sum_\lambda q^{|\lambda|}$ (sum over partitions with parts $\le m$) generates a *base partition*: an ordinary partition that slides underneath all the lattice paths without affecting their relative positions. Every CPP of height m contains this trivial base.

The factor $(q;q)_m$ in $h_m = (q;q)_m \cdot g_m$ *removes* the base partition via inclusion–exclusion. Expanding $\prod_{k=j+1}^{n}(1-q^k)$: each factor $(1-q^k)$ offers a binary choice — leave level k alone (coefficient 1) or subtract its base strip (coefficient $-q^k$). The subset U records which levels are being subtracted.

Two types of defect

The triple (S, U, π) carries two independent layers of sign:

	S **(crossings)**	U **(base removal)**
What	peeled layers	subtracted base strips
Range	$S = \{1, \ldots, j\}$	$U \subseteq \{j{+}1, \ldots, n\}$
NILP meaning	virtual crossings between detached and remaining paths	inclusion–exclusion for removing trivial padding
Sign	$(-1)^{\lvert S\rvert}$	$(-1)^{\lvert U\rvert}$
Weight	$\sigma(S) = 1 + \cdots + j$	$\sigma(U)$

The total sign is $(-1)^{|S|+|U|}$ and the total weight is $\sigma(S) + \sigma(U) + \mathrm{wt}(\pi)$. Fixed points of the involution must have $|S| + |U|$ even; the ideal fixed points are $(\emptyset, \emptyset, \pi)$ with sign $+1$.

5 The involution

To prove positivity of $Q_{n,c}(q)$, we need a sign-reversing, weight-preserving involution $\iota\colon \mathcal{G}_{n,c} \to \mathcal{G}_{n,c}$ whose fixed points all have positive sign.

The sign is $(-1)^{|S|+|U|}$, so ι must change the parity of $|S| + |U|$. There are three types of elementary moves.

5.1 Type 1: flip a U-element

Choose some $k \in \{j+1, \ldots, n\}$ and toggle its membership in U:

$$(S, U, \pi) \longmapsto (S, U \triangle \{k\}, \pi').$$

Here π does not change, so weight preservation requires $\sigma(U \triangle \{k\}) = \sigma(U) \pm k$ and $\mathrm{wt}(\pi')$ adjusts accordingly.

If $\pi' = \pi$ (the CPP is unchanged), then the weight changes by $\pm k$ from the U-part alone. This means a pure U-flip can only be weight-preserving if we also modify π.

5.2 Type 2: peel/glue

Extend or shrink the initial segment:

$$(S, U, \pi) \longmapsto (S', U', \pi'), \qquad |S'| = |S| \pm 1.$$

Peeling ($j \to j+1$): detach layer $j+1$ from π, producing a CPP π' of height $n-j-1$. The weight of the detached layer (horizontal strip at level $j+1$) must be absorbed.

Gluing ($j \to j-1$): attach a new outermost layer to π, producing π' of height $n-j+1$.

5.3 Type 3: transfer at the boundary

Move element $j+1$ between S and U:

- If $j+1 \in U$: remove $j+1$ from U and extend S to $[j+1]$. This increases $|S|$ by 1 and decreases $|U|$ by 1, so $|S|+|U|$ is unchanged — **no sign change**. Not useful for the involution directly, but useful as a preparatory step.
- If $j+1 \notin U$: extend S to $[j+1]$ and the parity of $|S| + |U|$ increases by 1. This *is* a sign-reversing move.

6 Comparison with the ALCD framework

	ALCD framework	**CPP framework**
Ground set	$(S, \mathfrak{d})$	(S, U, π)
S	$S \subseteq [n]$ (arbitrary)	$S = [j]$ (initial segment)
Combinatorial object	$\mathfrak{d} \in H_{m,c}$ (ALCD)	$\pi \in \mathrm{CPP}_{m,c}$ (CPP)
Requires	$h_m(q) \geq 0$	$g_m(q) \geq 0$ (automatic)
Sign	$(-1)^{\lvert S\rvert}$	$(-1)^{\lvert S\rvert+\lvert U\rvert}$
Weight	$\sigma(S) + \mathrm{wt}(\mathfrak{d})$	$\sigma(S) + \sigma(U) + \mathrm{wt}(\pi)$
NILP meaning	$\mathfrak{d}$ encodes (non)-intersecting paths	π is always non-intersecting
Involution	peel/glue at one level	three types of moves

The key advantage of the CPP framework is robustness: $g_m(q)$ is a nonneg polynomial by definition (it counts CPPs), so the ground set is well-defined without any additional assumptions.

The cost is an extra layer of structure: the subset U adds a second source of alternating signs. But this also provides more flexibility for the involution, since there are more elementary moves available.

7 Structure of the involution

The involution ι on $\mathcal{G}_{n,c}$ should proceed as follows.

1. **If $U \neq \emptyset$:** try to cancel using a U-flip. Let $k = \min(U)$ or $k = \max(U)$ (the canonical choice). If there exists π' with $\mathrm{wt}(\pi') = \mathrm{wt}(\pi) + k$ (or $\mathrm{wt}(\pi) - k$) and the same height, pair $(S, U, \pi) \leftrightarrow (S, U \setminus \{k\}, \pi')$.

2. **If $U = \emptyset$ and $j \geq 1$:** try to cancel using a peel/glue. The outermost layer of π at level j can be detached (if the strip size matches) or a new layer can be attached.

3. **If $U = \emptyset$ and $j = 0$:** the element $(S, U, \pi) = (\emptyset, \emptyset, \pi)$ has sign $+1$. This is a candidate **fixed point**. It contributes to $Q_{n,c}(q)$ if and only if no valid pairing exists.

The fixed points of ι are triples $(\emptyset, \emptyset, \pi)$ that survive all cancellation attempts. These are CPPs of height n that contribute $q^{\mathrm{wt}(\pi)}$ to $Q_{n,c}(q)$.

Remark 6. The fixed points are a *subset* of the CPPs of height n. Not every height-n CPP survives: some are paired with elements having $U \neq \emptyset$ or $S \neq \emptyset$. The surviving CPPs are exactly those that generate $Q_{n,c}(q)$.

8 Three strategies for the involution

The two signed sets S and U can be handled in several ways.

8.1 Strategy 1: sequential sieve

First cancel all U-signs (the $(q)_m$ inclusion–exclusion), then cancel all S-signs (the crossing cancellation). This is equivalent to:

1. Prove $h_m \in \mathbb{N}[q]$ (the U-sieve produces a nonneg result), then

2. apply the Franklin involution from CANONICAL_PEEL.tex to the residual.

This is the original approach. It works when $h_m \geq 0$, but fails for $d \equiv 0 \pmod 3$ because the U-sieve does not close up cleanly.

8.2 Strategy 2: interleaved sieve

At each step, choose whether to toggle an element of S or an element of U, based on a priority rule. The Franklin comparison generalises: compare the "cheapest" U-flip (smallest $k \in U$ or smallest $k \notin U$) with the cheapest peel/glue, and do whichever is smaller.

This interleaves the two types of cancellation, potentially resolving cases where the sequential approach gets stuck.

8.3 Strategy 3: combined set

Define $T = S \cup U \subseteq [n]$ and run a single Franklin sieve on T. The sign is $(-1)^{|T|}$ and the weight is $\sigma(T) + \mathrm{wt}(\pi)$. At each level k, toggle $k \in T$ (adding or removing k), adjusting the CPP height if k is at the S–U boundary.

The difficulty: toggling $k \in T$ may change whether k belongs to S or U, which changes the CPP height, which changes the set of available CPPs. The coupling between $|S|$ and the CPP height is the main obstruction.

9 The priority rule

The involution must choose which move to apply. We propose the following priority:

1. **U-cancellation first.** If $U \neq \emptyset$, work with U before touching S or π. The U-flip is the simplest move (it does not change the CPP height), so it should be preferred.
2. **Smallest element.** When cancelling via U, always target $k = \min(U)$ (analogous to the Franklin involution targeting the smallest part).
3. **Peel/glue as fallback.** Only modify S (and hence the CPP height) when no U-cancellation is available.

This priority ensures:

- Elements with $U \neq \emptyset$ are handled first, leaving only $(S, \emptyset, \pi)$ elements for the peel/glue step.
- The U-cancellation step is entirely within the set of CPPs of the *same height*, making it easier to verify involutivity.
- The peel/glue step operates on a smaller residual set, reducing the complexity of the canonical peel problem.

10 Open questions

1. **U-cancellation mechanism.** When $k \in U$, how exactly is the partner CPP π' with $\mathrm{wt}(\pi') = \mathrm{wt}(\pi) \pm k$ chosen? This requires a canonical bijection between weight-w and weight-$(w+k)$ CPPs of the same height, which exists when g_m has nondecreasing coefficients (up to a point) but needs to be made explicit.
2. **Interaction between U and peel/glue.** After the U-cancellation step eliminates all elements with $U \neq \emptyset$, the residual elements are $(S, \emptyset, \pi)$. These have sign $(-1)^{|S|}$ and weight $\sigma(S) + \mathrm{wt}(\pi)$. This is exactly the ALCD framework restricted to initial-segment subsets S. The peel/glue involution from CANONICAL_PEEL.tex applies to this residual.
3. **Does the U-step close up?** For the U-cancellation to define an involution on its own, we need: if (S, U, π) maps to (S, U', π'), then applying the same rule to (S, U', π') returns to (S, U, π). This requires the bijection $\pi \leftrightarrow \pi'$ to be its own inverse.
4. **Weight constraint.** The U-flip at level k changes the U-weight by $\pm k$. For weight preservation, $\mathrm{wt}(\pi)$ must change by $\mp k$. This means we need to find, for each CPP π of height m, a partner π' of the same height with $\mathrm{wt}(\pi') = \mathrm{wt}(\pi) \pm k$. This is a matching problem on CPPs — nontrivial but concrete.

5. **Fixed points of the U-step.** Elements (S, U, π) with $U \neq \emptyset$ that cannot be U-cancelled (no valid π' exists) must be handled by a different mechanism. Do these exist? If so, they need to be routed to the peel/glue step.

11 Connection to previous notes

- **CANONICAL_PEEL.tex:** The peel/glue framework for the ALCD ground set. Sections 3–4 (Franklin comparison, weight condition) apply directly to the residual after U-cancellation.
- **Q_NILP.tex:** The enlarged path space and crossing framework. Section 4.3 (intersecting paths as relaxed interlacing) describes what happens when $S \neq \emptyset$.
- **Q_LOCAL.tex:** The local rule and conflict detection ($A \cap B$). Relevant for the peel/glue step on the residual.